

	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	------	------------------	-----------------------

1 PURPOSE

This document describes ASTM format for patient results sending from an AL22 towards a LIS and receipt of orders from a LIS through RS or NET.

2 APPROBATION AND HISTORY

Revision	Date	Author	Raison et nature de la modification	Approbation
01	10/01/11	V.Heaney	Document creation	J. Stawicki
02	10/02/11	V.Heaney	'Update following V2.0.0 corrections.	J. Stawicki

3 DIFFUSION

4 DEFINITIONS AND ABBREVIATIONS

- **N/A** : Not applicable
- **Sender** : Appliance in charge of sending data towards another appliance
- **Receiver**: Appliance in charge of accepting and receiving incoming data from a sender
- **LIS**: Laboratory Instrument system, the host.
- **Order**: A request for analysis incoming from a LIS and concerning one patient.

5 REFERENCES

6 SYSTEM OVERVIEW

The Mythic22 AL is the auto-loader version of the Mythic 22, haematology 5DIF analyzer.

	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	-------------	-------------------------	-------------------------------

1	PURPOSE	1
2	APPROBATION AND HISTORY	1
3	DIFFUSION.....	1
4	DEFINITIONS AND ABBREVIATIONS	1
5	REFERENCES	1
6	SYSTEM OVERVIEW	1
7	GENERAL SPÉCIFICATIONS.....	3
7.1	FORMAT OVERVIEW	3
7.2	ASTM FORMAT	3
7.2.1.1	Description.....	3
7.2.1.2	Unidirectional communication diagram for AL22	3
7.2.1.3	Bidirectional communication diagram for AL22.....	5
7.2.2	ASTM Fields Implementation	7
7.2.2.1	Header H Record	7
7.2.2.2	Patient P Record	7
7.2.2.3	Order O Record	8
7.2.2.4	Result R Record.....	9
7.2.2.5	Manufacturer Information M Record.....	9
7.2.2.6	Terminator T Record	10

	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	------	------------------	-----------------------

7 GENERAL SPÉCIFICATIONS

7.1 Format overview

ASTM format is used for standardized communication with a LIS. This note describes the way ASTM fields are filled and numbered in order to allow third party developers to design their own software solution. ASTM used version is referenced as standard **E1394**. The low-layer protocol referenced as standard **E1381** will not be discussed in this document.

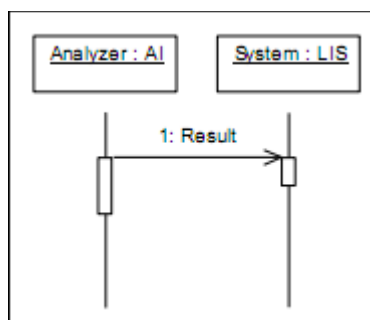
7.2 ASTM Format

7.2.1.1 Description

This is an ASCII protocol based on handshakes. Two modes are internally managed. The unidirectional mode and the bidirectional mode (which contains several sub-modes)

- UNIDIRECTIONAL Mode allows the AL22 to send patient results towards a LIS. (either results read from the archive then sent towards the LIS or directly sent afterwards an analysis)
- BIDIRECTIONAL Mode is intended to receive a worklist from a LIS, apply the required tests and send results back to the LIS. In the case of the AL22, worklist orders received from a LIS will be processed to enqueue the current worklist of the day. Results are sent back after analysis completion.

7.2.1.2 Unidirectional communication diagram for AL22



Patient result sending for DIF where SID is TEST SID 1 and realized on February, the 11th. Below, the 22 parameters transmission.

```

H|\^&|||ORPHEE AL22||||LIS_ID||P|E 1394-97|20110211033633
P|1|TEST PID 1|TEST SID 1|34|TEST ID||19900101|M||||HOUSE|comment|||||||NY|||||
O|1|TEST SID 1|34|^^^DIF|R||20000|||||STANDARD||||00000|^||||F||||
R|1|^^^WBC|0 |10e2/uL|4 to 12 |LL||F|||20110211033633|
R|2|^^^RBC|0 |10e4/uL|4 to 6 |LL||F|||20110211033633|
R|3|^^^HGB|.....|g/dL|11.0 to 17.0||F|||20110211033633|
R|4|^^^HCT|0.0 |%|35.0 to 55.0|LL||F|||20110211033633|
R|5|^^^PLT|0.0 |10e4/uL|150.0 to 400.0|LL||F|||20110211033633|
R|6|^^^LYM|0 |10e3/uL|1 to 5 |LL||F|||20110211033633|
R|7|^^^MON|0 |10e3/uL|0 to 1 |L||F|||20110211033633|
R|8|^^^NEU|0 |10e3/uL|2 to 8 |LL||F|||20110211033633|
R|9|^^^LYM%|0.0 |%|25.0 to 50.0|LL||F|||20110211033633|

```

	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	------	------------------	-----------------------

```

R|10|^^^MON%|0.0 |%|2.0 to 10.0|LL||F|||20110211033633|
R|11|^^^NEU%|0.0 |%|50.0 to 80.0|LL||F|||20110211033633|
R|12|^^^MCV|.....|fL|80.0 to 100.0||F|||20110211033633|
R|13|^^^MCH|.....|pg|26.0 to 34.0 ||F|||20110211033633|
R|14|^^^MCHC|.....|g/dL|31.0 to 35.5 ||F|||20110211033633|
R|15|^^^RDW|.....|%|10.0 to 16.0 ||F|||20110211033633|
R|16|^^^MPV|.....|fL|7.0 to 11.0 ||F|||20110211033633|
R|17|^^^PCT|.....|%|0.200 to 0.500||F|||20110211033633|
R|18|^^^PDW|.....|%|10.0 to 18.0 ||F|||20110211033633|
R|19|^^^EOS|0 |10e3/uL|0 to 0 |N||F|||20110211033633|
R|20|^^^BAS|0 |10e3/uL|0 to 0 |N||F|||20110211033633|
R|21|^^^EOS%|0.0 |%|0.0 to 5.0 |N||F|||20110211033633|
R|22|^^^BAS%|0.0 |%|0.0 to 2.0 |N||F|||20110211033633|
M|1|Alert=00011010000000|Flag=0000000000000000

```

L|1|N

Patient result sending for CBC where SID is TEST SID 10 and realized on February, the 10th.
Below, the 18 parameters transmission.

```

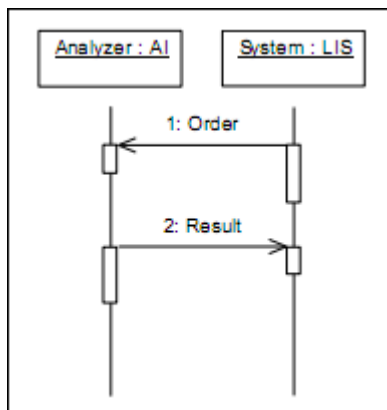
H|\^&|||ORPHEE AL22|LIS_ID|P|E 1394-97|20110210071116
P|1|TEST PID 10|TEST SID 10|33|TEST ID||19420422|M|||HOUSE|comment|CALIFORNIE|
O|1|TEST SID 10|33|^^^CBC|R||00000|STANDARD||00000|^||F|||
R|1|^^^WBC|0 |10e2/uL|4 to 12 |LL||F|||20110210071116|
R|2|^^^RBC|0 |10e4/uL|4 to 6 |LL||F|||20110210071116|
R|3|^^^HGB|.....|g/dL|11.0 to 17.0||F|||20110210071116|
R|4|^^^HCT|0.0 |%|35.0 to 55.0|LL||F|||20110210071116|
R|5|^^^PLT|0.0 |10e4/uL|150.0 to 400.0|LL||F|||20110210071116|
R|6|^^^MCV|.....|fL|80.0 to 100.0||F|||20110210071116|
R|7|^^^MCH|.....|pg|26.0 to 34.0 ||F|||20110210071116|
R|8|^^^MCHC|.....|g/dL|31.0 to 35.5 ||F|||20110210071116|
R|9|^^^RDW|.....|%|10.0 to 16.0 ||F|||20110210071116|
R|10|^^^MPV|.....|fL|7.0 to 11.0 ||F|||20110210071116|
R|11|^^^PCT|.....|%|0.200 to 0.500||F|||20110210071116|
R|12|^^^PDW|.....|%|10.0 to 18.0 ||F|||20110210071116|
M|1|Alert=00001001000000|Flag=0000000000000000

```

L|1|N

	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	------	------------------	-----------------------

7.2.1.3 Bidirectional communication diagram for AL22



Data patient receipt which are translated into patient orders to populate the current worklist.
For instance, the second order named TEST SID 2, born on March the 12th in 1985, requires a CBC analysis by Dr House in Oregon.

H|\^&|||ORPHEE AL22||||LIS_ID||P|E 1394-97|20110007182836

P|1|TEST PID 1|TEST SID 1|TEST ID||19900101|M||||HOUSE|comment| OREGON |||||

O|1|TEST SID 1||^DIF|R||20000|STANDARD|||03012|^|

P|1|TEST PID 2|TEST SID 2|TEST ID||19850312|F||||HOUSE|comment| OREGON |||||

O|1|TEST SID 2||^CBC|R||00000|STANDARD|||03121|^|

P|1|TEST PID 3|TEST SID 3|TEST ID||20020613|M||||HOUSE|comment| OREGON |||||

O|1|TEST SID 3||^DIF|R||10000|STANDARD|||03133|^|

P|1|TEST PID 4|TEST SID 4|TEST ID||19870805|U||||HOUSE|comment| OREGON |||||

O|1|TEST SID 4||^CBC|R||00000|STANDARD|||03142|^|

P|1|TEST PID 5|TEST SID 5|TEST ID||19751007|M||||HOUSE|comment| OREGON |||||

O|1|TEST SID 5||^CBC|R||20000|STANDARD|||03153|^|

P|1|TEST PID 6|TEST SID 6|TEST ID||19721110|F||||HOUSE|comment| OREGON |||||

O|1|TEST SID 6||^DIF|R||20000|STANDARD|||03163|^|

P|1|TEST PID 7|TEST SID 7|TEST ID||19621206|M||||HOUSE|comment| OREGON |||||

O|1|TEST SID 7||^CBC|R||10000|STANDARD|||03171|^|

P|1|TEST PID 8|TEST SID 8|TEST ID||19481005|U||||HOUSE|comment| OREGON |||||

O|1|TEST SID 8||^CBC|R||00000|STANDARD|||03185|^|

P|1|TEST PID 9|TEST SID 9|TEST ID||19440504|M||||HOUSE|comment| OREGON |||||

O|1|TEST SID 9||^DIF|R||20000|STANDARD|||03194|^|

P|1|TEST PID 10|TEST SID 10|TEST ID||19420422|M||||HOUSE|comment| OREGON |||||

	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	------	------------------	-----------------------

O|1|TEST SID 10||^^^CBC|R||00000|||||||STANDARD|||||03025|^|||||||

L|1|N

During a receipt, errors may occur :

ERR_ASTM_CRC : CRC error. The received astm frame is contains error(s), CRC is wrong

ERR_ASTM_PARAM : Number of received parameters for a frame type does not match the standard requirements.

ERR_ASTM_WRONG_FRM : The received ASTM frame was not expected by the AL22.

ERR_ASTM_WRONG_VERSION : ASTM version used by the LIS is not recognized by the AL22. It must be ***E1394-97***.

ERR_ASTM_USE_DETECTED : Errors occur during ASTM data processing. These errors do not depend of an ASTM encapsulation. This might be a received order already available in the current worklist, for instance.

	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	------	------------------	-----------------------

7.2.2 ASTM Fields Implementation

7.2.2.1 Header H Record

Field No.	Label	Static Content	Description
H1	Type ID	H	Header Type
H2	Delimiter Definit.	\^&	fields delimiter
H5	Sender Name/ID	ORPHEE AL22	Name of the sender equipment (automaton)
H10	Received ID	LIS_ID	Name of the LIS
H11	Comment	-	-
H12	Processing ID	P	Process type (patient data)
H13	Version No.	E1394-97	-
H14	Date and Time msg	-	Format: YYYYMMDDHHMMSS

Example:

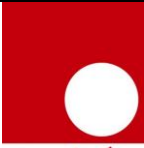
H|\^&|||ORPHEE AL22|LIS_ID||P|E 1394-97|20110211033633

7.2.2.2 Patient P Record

Field No.	Label	Static Content	Description
P1	type ID	P	-
P2	Sequence No.	-	Record number of the patient, ex : 1, 2..etc
P3	Practiced As. Pat. No.	-	Practiced Assigned Patient Number : AL22 PID
P4	Lab. As. Pat. ID	-	Laboratory Assigned Patient ID : AL22 SID
P5	Patient ID No3	-	AL22 Sequence Number
P6	Patient Name	-	AL22 ID
P8	Birthdate	-	Format : YYYYMMDD
P9	Patient Sex	-	M(ale) F(emale) or U(ndefined)
P14	Attending Phys. ID	-	ID of the physician.
P15	Special Field No.1	-	Comment related to the patient
P26	Location	-	Location of the examination

Example:

P|1|TEST PID 1|TEST SID 1|TEST ID||19900101|M|HOUSE|comment| OREGON

 Orphée	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	------	------------------	-----------------------

7.2.2.3 Order O Record

On ajoute les informations de rack tels que le numéro et type de rack et la position dans le rack.

Field No.	Label	<u>Static Content</u>	<u>Description</u>
O1	Record Type ID	O	-
O2	ASTM Seq. Number	1	-
O3	Specimen ID		AL22 SID
O4	Instrument Spec. ID		sequence number assigned by AL22
O5	Universal Test ID	^^^xxx	where xxx = CBC or DIF
O6	Priority	R	-
O8	Spec. Collec Date/Time		Draw Date.: DHHMM (D:0,1,2→unknown/today/yesterday)
O16	Specimen Type	STANDARD	-
O21	Lab. Field. N°1		Rack pos, format : TTNNP (type, number, position)
O22	Lab. Field. N°2	.	M (match) ou/et R (rerun) separated by cmpnt delimiters
O26	Report Type	F ou X	F if successful, X if errors occur.

Example:

O|1|TEST SID 1|34|^^^DIF|R||20000| |||||STANDARD| ||||00000|^| |||F| |||

	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	------	------------------	-----------------------

7.2.2.4 Result R Record

N° champ	Libellé	Contenu statique	Description
R1	Record Type ID	R	-
R2	Record Seq. Number	-	1...n
R3	Universal Test ID	^^^xxx	Where xxx may be WBC, RBC, HGB..and so on..
R4	Measurement Value	-	Measured value ex:10.9
R5	Unité	-	Unit, ex : K/uL
R6	Reference Range	-	Range of values : ex : 3.0 to 12.0
R7	Result Flags	L, H, LL, HH, A, N	Alarms, A → Dil, N→Normal
R9	Result Status	F ou X	F if successful, X if errors occur
R13	Date/Time test completed	-	Format: YYYYMMDDHHMMSS

Example:

R|1|^^^WBC|0 |10e2/uL|4 to 12 |LL||F|||20110211033633|

7.2.2.5 Manufacturer Information M Record

This record is used in order to transmit alerts and flags towards the LIS.

Field No.	Label	Static Content	Description
M1	<u>Record Type ID</u>	M	
M2	<u>Sequence Number</u>	1	
M3	<u>Instrument Alert</u>	-	Bytes: 1-6 Alert= 7 1 = W_CL 8 1 = R_CL 9 1 = O_CL 10 1 = O_DF 11 1 = SU_ND 12 1 = SU_F 13 1 = QC_ND 14 1 = QC_F 15-20 Non-utilisés
M4	<u>Flag de test</u>	-	Bytes: 1-5 Flag= 6 1 = L1 7 1 = L5 8 1 = N1 9 1 = N2 10 1 = RL 11 1 = HL 12 1 = NL

	AL22	AL22 ASTM format	SEIS_FormatASTM 02
--	------	------------------	-----------------------

			13 1 = NH 14 1 = IC 15 1 = R1 16 1 = R2 17 1 = P1 18 1 = P2 19 1 = P3 20 non-utilisé.
--	--	--	--

Example:

M|1|Alert=00000101000000|Flag=0000000000000000

7.2.2.6 Terminator T Record

L1: Record Type ID, ici L.

L2: Sequence Number.
• *This field remains at 1.*

L3: Termination Code:
• *_This field remains at N.*

Example:

L|1|N